

FinTrust Digital Bank
Week 1- Data Analytics Report

AnalystLab Africa
Experience Lab Internship Programme

Data Analytics Track

FinTrust Financial Intelligence & Digital Banking

Support Solution

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GENERAL WEEK 1 REQUIREMENTS

A. Understand the Business Problem

1. What problem is FinTrust trying to solve?

FinTrust Digital Bank wants to better understand its customers and transaction activities and use data to support business, operational and risk-related decisions.

2. Why is this important?

Understanding customer behaviour and transaction patterns can help FinTrust identify areas for improvement, monitor transaction outcomes, understand channel usage and support better decision-making.

3. How can Data Analytics contribute?

Data Analytics can be used to profile the available data, identify patterns, calculate key performance indicators (KPIs), answer business questions and present findings through dashboards and reports.

4. What output could the Data Analytics track provide?

The track can provide a business intelligence report/dashboard showing customer behaviour, transaction activity, transaction values, channels, statuses and risk-review patterns.

B. Review Relevant Resources

The main resources relevant to the Data Analytics track are:

Resource	What it contains	How it will be used	Limitation
FinTrust Customer Data	Customer information such as segment, account type, age, city and account status	To understand customer characteristics and connect customers to their transactions	The data is synthetic and may not represent real customer behaviour
FinTrust Transaction Data	Transaction details including amount, type, channel, status and risk-review flag	To analyse transaction activity, value, channels, outcomes and risk-related patterns	Some Device_Type and Location values are missing

FinTrust Data Dictionary	Definitions, data types and business uses of the fields	To correctly interpret and validate the datasets	Analysis depends on the definitions provided
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C. Define Track Objectives

The main objectives are to:

1. Profile and validate the customer and transaction datasets.
2. Identify useful patterns in customer and transaction behaviour.
3. Develop meaningful analytical questions and KPIs.
4. Plan a dashboard that clearly communicates important findings.
5. Provide data-driven insights that can support FinTrust's decisions.

D. Identify Success Criteria

The solution will be considered successful if:

- The datasets are correctly understood and their relationship is established.
- The selected analytical questions can be answered using the available data.
- KPIs are clearly defined and accurately calculated.
- The dashboard communicates important patterns clearly.
- The findings are accurate, understandable and useful for management.

E. Initial Plan

Week 2: Clean and prepare the datasets, join the customer and transaction data, perform exploratory analysis and calculate the selected KPIs.

Week 3: Build the Power BI dashboard, create visuals and filters, and analyse the major patterns identified.

Week 4: Validate the results, refine the dashboard, document key insights and prepare the final submission.

2. DATA ANALYTICS TRACK

Part A -Business Understanding

Business Questions

Management should be able to answer questions such as:

1. Which customer segments are the most active?
2. Which transaction types and channels are used most?
3. What transaction types and channels contribute the most transaction value?
4. How do transaction statuses vary across channels and transaction types?
5. What patterns are visible in transactions marked for risk review?
6. How does transaction activity change over time?

Decisions Supported by Data Analysis

The analysis can support decisions relating to customer engagement, digital channel performance, transaction processing issues, operational monitoring and areas requiring further risk review.

Key Stakeholders

- Management and executives
- Operations team
- Customer experience team
- Risk and compliance team
- Product and digital banking teams
- Data and analytics team

3. DATA UNDERSTANDING

Reviewing both customer and transaction dataset

Customer Dataset

- **Records:** 1,500
- **Columns:** 12
- **Field names:** Customer_ID, Customer_Name, Age, Gender, City, Customer_Segment, Account_Type, Tenure_Months, Digital_Engagement_Score, Monthly_Income_Band, Preferred_Channel, Account_Status
- **Numerical variables:** Age, Tenure_Months, Digital_Engagement_Score
- **Categorical variables:** Customer_ID, Customer_Name, Gender, City, Customer_Segment, Account_Type, Monthly_Income_Band, Preferred_Channel, Account_Status
- **Date/Time variables:** None

- **Missing values:** None

Transaction Dataset

- **Records:** 12,000
- **Columns:** 11
- **Fields:** Transaction_ID, Customer_ID, Transaction_DateTime, Transaction_Type, Amount_NGN, Channel, Device_Type, Location, International_Transaction, Transaction_Status, Risk_Review_Flag
- **Numerical variables:** Amount_NGN
- **Categorical variables:** Transaction_ID, Customer_ID, Transaction_Type, Channel, Device_Type, Location, International_Transaction, Transaction_Status, Risk_Review_Flag
- **Date/Time variable:** Transaction_DateTime
- **Missing values:** Device_Type — 96 records; Location — 96 records
- **Data period:** January 1, 2026 to March 31, 2026

Relationship Between the Datasets

The datasets are connected through **Customer_ID**. One customer can have multiple transactions, making this a one-to-many relationship.

Initial Data-Quality Observations

There are no completely duplicated transaction records or duplicate Transaction_IDs. All transaction Customer_IDs match the customer dataset. The main issues observed at this stage are the missing Device_Type and Location values.

4. ANALYTICAL QUESTIONS

The following questions will guide the analysis:

1. How does transaction volume vary across customer segments?
2. Which transaction types are most frequent and which generate the highest transaction value?
3. Which transaction channels are most used?
4. How does average transaction value vary by customer segment and channel?
5. How do successful, failed, pending and reversed transactions vary by transaction type and channel?
6. How do international and domestic transactions differ in volume, value and status?

7. What patterns are associated with Risk_Review_Flag across channels, transaction types, locations and customer segments?
8. How does transaction activity change from January to March 2026?

5. KPI DEFINITION TABLE

KPI	Definition	Why It Matters	Required Data
Total Transactions	Count of all transactions	Measures overall transaction activity	Transaction_ID
Total Transaction Value	Sum of transaction amounts	Shows the total value processed	Amount_NGN
Average Transaction Value	Total transaction value ÷ total transactions	Shows the average value per transaction	Amount_NGN, Transaction_ID
Successful Transaction Rate	Successful transactions ÷ total transactions × 100	Measures transaction success	Transaction_Status, Transaction_ID
Failed Transaction Rate	Failed transactions ÷ total transactions × 100	Helps monitor transaction failures	Transaction_Status, Transaction_ID
Risk Review Rate	Risk-reviewed transactions ÷ total transactions × 100	Shows the proportion of transactions flagged for review	Risk_Review_Flag, Transaction_ID
Active Customer Rate	Active customers ÷ total customers × 100	Shows the proportion of customers with an active account status	Account_Status, Customer_ID
Transactions per Customer	Total transactions ÷ unique customers	Shows average transaction activity per customer	Transaction_ID, Customer_ID

6. DASHBOARD PLANNING

Proposed Dashboard Wireframe

Header:

FinTrust Digital Bank — Financial Intelligence Dashboard

KPI Cards:

Total Transactions | Total Transaction Value | Average Transaction Value | Success Rate | Failed Rate | Risk Review Rate

Section 1 — Transaction Overview

- Transaction volume over time
- Transaction volume by transaction type

Section 2 — Customer Behaviour

- Transaction activity by customer segment
- Transaction value by customer segment

Section 3 — Channel & Status

- Transaction volume by channel
- Transaction status by channel/type

Section 4 — Risk Review

- Risk-review transactions by channel
- Risk-review transactions by transaction type/location

Filters:

- Date
- Customer Segment
- Transaction Type
- Channel
- Location
- Transaction Status
- International Transaction
- Risk Review Flag

Expected Insights:

The dashboard should help identify the most active customer segments, major transaction types and channels, transaction-status patterns, changes in activity over time and areas with higher levels of risk-review flags.

7. INITIAL ANALYSIS PLAN

The analysis will begin with data validation and preparation, followed by exploratory analysis of customers and transactions. The customer and transaction datasets will then be connected using Customer_ID.

The selected KPIs and analytical questions will be analysed using appropriate tools such as Excel, Python/Pandas and SQL. Power BI will be used to present the results in an interactive dashboard.

The analysis will focus on producing clear, evidence-based insights that can support FinTrust's business and operational decisions.

Dashboard Wireframe

The proposed dashboard is designed to provide a clear overview of FinTrust’s customer and transaction performance. It includes KPI cards, transaction and customer analysis, channel and status monitoring, risk-review analysis, filters, and expected business insights.

FinTrust Digital Bank - Financial Intelligence Dashboard

Filters: Date Customer Segment Transaction Type Channel Location Status International Risk Flag					
Total Transactions KPI	Total Value KPI	Avg Transaction KPI	Success Rate KPI	Failed Rate KPI	Risk Review Rate KPI
1. Transaction Overview - Line chart: transaction volume over time - Bar chart: transactions by transaction type		2. Customer Behaviour - Bar chart: activity by customer segment - Bar chart: value by customer segment			
3. Channel & Status - Bar chart: transactions by channel - Stacked chart: status by channel/type		4. Risk Review - Bar chart: risk review by channel/type - Bar chart: risk review by location			

Expected insights: customer activity | channel usage | transaction outcomes | time trends | risk-review patterns